

Proposed Solution

Date	30 June 2026
Team ID	xxxxxxxxxx
Project Title	Rising Waters - Automated IoT Flash Flood Prediction and Warning System
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform disaster response and flash flood management using machine learning, boosting emergency dispatch efficiency and forecast accuracy. It tackles slow manual data tracking inefficiencies, promising better automated operations, reduced risks to human lives, and safer local communities. Key features include a machine learning-based river level predictive model and real-time citizen warning systems.

Project Overview	
Objective	The primary objective is to revolutionize the disaster response process by implementing advanced machine learning techniques, ensuring faster and more accurate flash flood assessments.
Scope	The project comprehensively assesses and enhances the flood monitoring process, incorporating machine learning for a more robust and efficient automated early-warning system.
Problem Statement	
Description	Addressing inaccuracies and time lag inefficiencies in the current manual flood tracking and weather analysis systems adversely affects operational dispatch speed and citizen safety.
Impact	Solving these issues will result in improved emergency operational efficiency, reduced casualty risks, and an overall enhancement in the evacuation coordination process, contributing to public safety and community resilience.
Proposed Solution	
Approach	Employing machine learning techniques to analyze and predict flash flood probabilities, creating a dynamic and adaptable live emergency broadcast and warning system.
Key Features	· Implementation of a machine learning-based precipitation and river-gauge assessment model. · Real-time decision-making for quicker 4 to 6-hour advance emergency alerts. · Continuous learning to adapt to evolving climate shifts and

	extreme storm anomalies.
--	--------------------------

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU- High-performance parallel processing for training Python ML storm data models.
Memory	RAM specifications	8 GB-Sufficient RAM allocation for real-time data ingestion pipelines and concurrent API calls.
Storage	Disk space for data, models, and logs	1 TB SSD-Disk space for historical regional weather databases, model parameters, and active sensor logs.
Software		
Frameworks	Python frameworks	Flask-Python-based lightweight backend frameworks to host prediction model endpoints
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn-Core data manipulation, mathematical processing, and flood risk visualization utilities.
Development Environment	IDE	Jupyter Notebook, pycharm IDE tools for rapid machine learning testing and automated script execution.
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv-Government Meteorological

		Department historical records, IoT live river-gauge telemetry streams, CSV/JSON format.
--	--	---